

```
# Loop in Python ---->
# (1). while Loop
# (2). For Loop
```

```
# (1). while Loop ---->

# initialization
# condition
# Increment or decrement
```

```
# Q.1 Write a program who print from 1 to 3 ?
```

```
i=1 ##Initialize
n=3
while(i<=n): ## Condition
    print(i)
    i=i+1    ###Increment
```

```
1
2
3
```

```
# # step-1 (i=1,n=3)
# while(1<=3):
#     print(1)
#     i = 1 + 1 = 2

# # step-2 (i=2,n=3)
# while(2<=3):
#     print(2)
#     i=2+1=3

# # step-3 (i=3,n=3)
# while(3<=3):
#     print(3)
#     i = 3 + 1 = 4

# # step-4 (i=4,n=3)
# while(4<=3):
#     break

# Final Output ----> 1,2,3
```

```
# Q.2 Write a user defined program who will take starting and last number from user and print all values.
```

```
i = int(input("Enter first number:"))
n = int(input("Enter last number:"))
p = int(input("Enter step value:"))
while(i<=n):
    print(i)
    i = i + p
```

```
Enter first number:1
Enter last number:10
Enter step value:3
1
4
7
10
```

```
# Q.3 Write a program who print from 5 to 1 and step value will be 2 .
```

```
# Q.4 Write a program who calculate sum of a given number's range .
```

```
i = 1
n = 4
sum = 0
while(i<=n):
    sum = sum + i
    i = i + 1
print(sum)
```

```
10
```

```
# 1+2+3+4 = 10
```

```
# # step-1 (i=1,n=4,sum=0)
# while(1<=4):
#     sum = 0 + 1 = 1
#     i = 1 + 1 = 2

# # step-2 (i=2,n=4,sum=1)
# while(2<=4):
#     sum=1 + 2 = 3
#     i = 2 + 1 = 3

# # step-3 (i=3,n=4,sum=3)
# while(3<=4):
#     sum= 3 + 3 = 6
#     i = 3 + 1 = 4

# # step-4 (i=4,n=4,sum=6)
# while(4<=4):
#     sum = 6 + 4 = 10
#     i = 4 + 1 = 5

# # step-5 (i=5,n=4,sum=10)
# while(5<=4):
#     break

# # Final Output ---> sum = 10
```

Q.5 Write a program who calculate sum of square of a given number ?

```
i=1
n=3
sum=0
while(i<=n):
    sum = sum + i**2
    i = i + 1
print(sum)
```

14

```
# 1**2 + 2**2 + 3**2 = 1+4+9 = 14
```

```
# # step-1 (i=1,n=3,sum=0)
# while(1<=3):
#     sum=0+1**2 = 1
#     i = 1+1 = 2

# # step-2 (i=2,n=3,sum=1)
# while(2<=3):
#     sum=1+2**2 = 1+4=5
#     i=2+1=3

# # step-3 (i=3,n=3,sum=5)
# while(3<=3):
#     sum = 5 + 3**2 = 5 + 9 = 14
#     i = 3+1 = 4

# # step-4 (i=4,n=3,sum=14)
# while(4<=3):
#     break

# Final Sum = 14
```

Q.5 sum of cube in a given range . user defined ??

```
# 5**3 = 5*5*5
```

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